

## Fall 2026A ASTRONOMY 302 Observing Proposal (10 points total)

**Abstract of Scientific Justification (300 words max) (2 points):** Summarize your program. Why is this observing program scientifically interesting? What target(s) are you trying to observe and with what observational modes? What science will your program uncover?

Jupiter rotates over a period of ~12 hours which causes a change in brightness over time that roughly follows a sinusoidal function. This change in brightness is due to the weather and storms in the atmosphere.

Brown Dwarfs are substellar objects that are not massive, or hot enough to fuse hydrogen into helium. Brown Dwarfs likely have similar changes in brightness over time due to similar circumstances as that of Jupiter.

The goal of this project is to determine the change in brightness over time of source 2MASSW J0036159+182110, a L3.5 brown dwarf and determine if this source exhibits a similar sinusoidal change in brightness over time. To do this we will use the Bok 90" telescope on Kitt Peak and the 90 Prime instrument to collect data. We will expose for 200s at a time attempting to get a signal-to-noise ratio between 50-200. This will allow us to analyze the cloud cover and temperature further expanding on ideas about how these factors change over time.

**Telescope Requested (0.5 points):** Bok 90"

**Instrument (0.5 points):** 90 Prime

**Tolerable Moon Brightness (dark, grey, bright, etc) (0.5 points):** Write out what moon brightness is acceptable for your program and why. How does the moon impact the sensitivity of your observations?

For good observations the moon will need to be dark. Brown Dwarfs are usually quite dim so a bright moon would make observing quite difficult to gather good data.

### Target List (2 points)

List potential targets you may observe and their basic properties like brightness. Under visibility write what months/days the target is up. For example, "Object can be observed Feb-March."

| Object Name | Right Ascension (deg) | Declination (deg) | V magnitude (i-band and r-band) | Object Type | Visibility |
|-------------|-----------------------|-------------------|---------------------------------|-------------|------------|
|             |                       |                   |                                 |             |            |

|                               |         |         |                    |      |                                                                                            |
|-------------------------------|---------|---------|--------------------|------|--------------------------------------------------------------------------------------------|
| 2MASSW<br>J0036159+1<br>82110 | 9.0671  | 18.3529 | 17.1922<br>19.5066 | L3.5 | Can be observed late<br>August through October<br><br><b>On September 30th:</b><br>~ 8.67h |
| 2MASSW<br>J0149090+2<br>95613 | 27.2875 | 29.9368 | 17.7745<br>19.9889 | M9.5 | Can be observed<br>September through<br>November<br><br><b>On September 30th:</b><br>~8h   |

### Experimental Design & Technical Description (4.5 points total)

How much time is required for your program and what signal-to-noise is necessary for your observations? (2 points)

Similar to Heinze et al. (2015) where they examined various brown dwarfs with the Kitt Peak 2.1m telescope using an F814W filter, we are aiming for a similar exposure time of ~200 seconds. This will hopefully yield a SNR between 60-200 shooting for ~100 for good observations. We will need to observe for as long as possible, hopefully most of the time that the source is up ~9 hours. This amount of time is necessary due to the nature of the project and the tight constraints for good observations.

How will these observations contribute toward the accomplishment of the goals outlined in the abstract? (2 points)

These observations will measure the brightness of the source over a period of (hopefully) 9 hours with these observations we can compare the values and plot how they change over time. With this we will ideally be able to see the sinusoidal function discussed in the abstract.

Back Up Plan. Describe a potential project that can be done with poor weather or archival data. (0.5 points)

Due to the necessity for clear skies and a dark moon for successful observations our backup plan if there is poor weather is to observe a brighter source such as an M dwarf star which also can have fluctuations in brightness due to starspots on the surface.